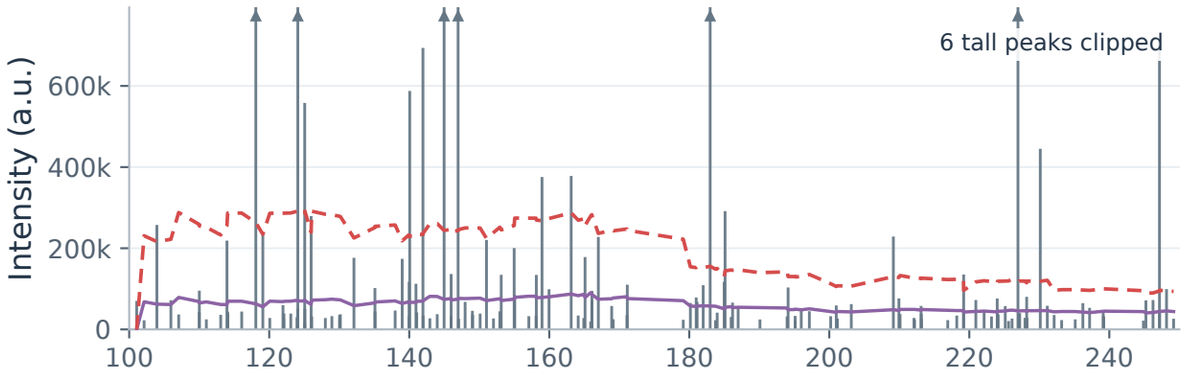
Adaptive MS1 filtering: real before / after

NIST_POS_Samp_01-01.mzXML • scan 1823 • RT 299.839 s • m/z 100–250

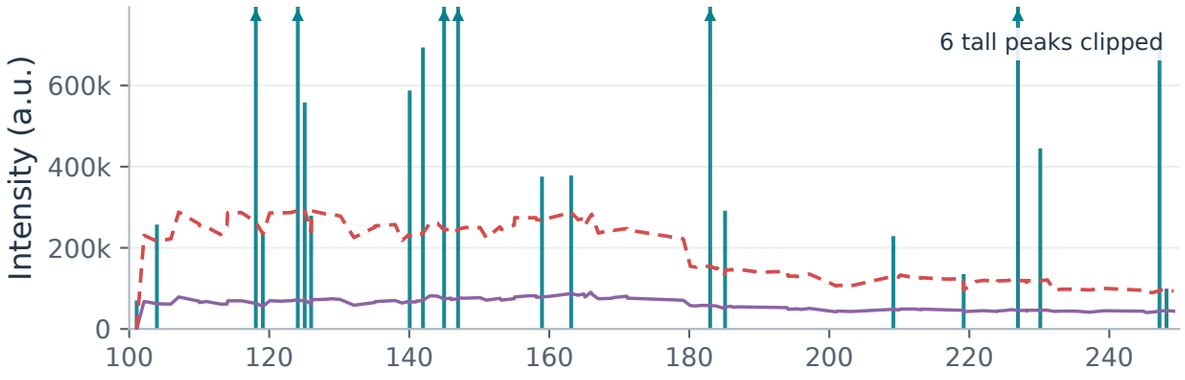
Cutoff view

Median default cutoff = 226.7k; 29% of axis height

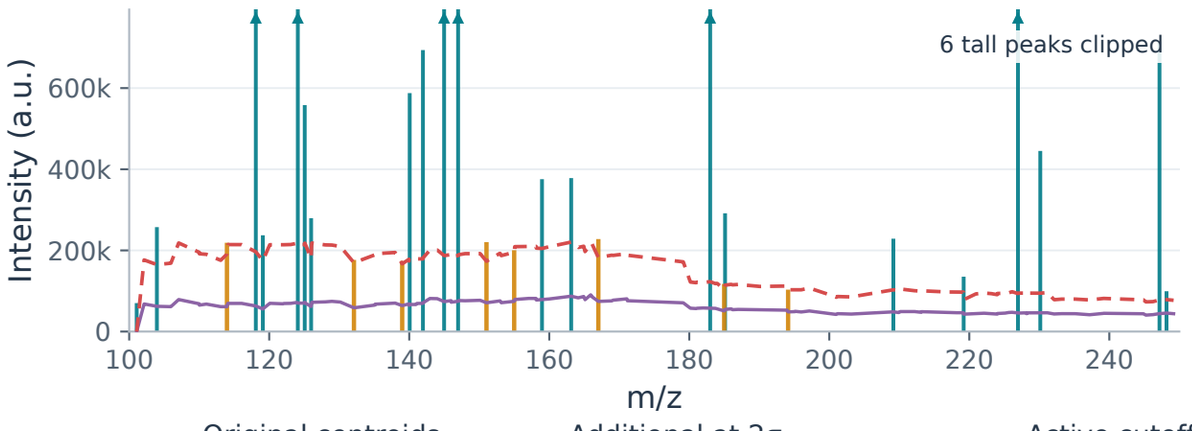
Before filtering · 148 centroids



Default 3σ · 21/148 kept (14.2%)



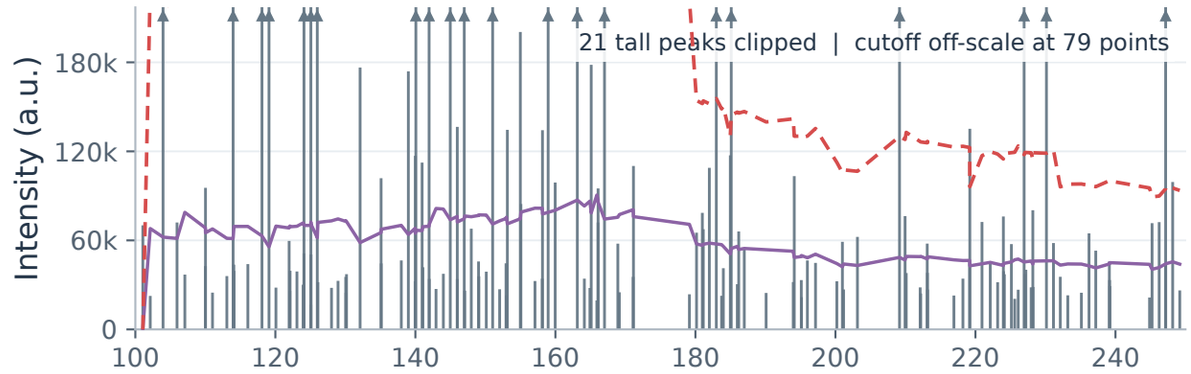
Less strict 2σ · 29/148 kept (+8)



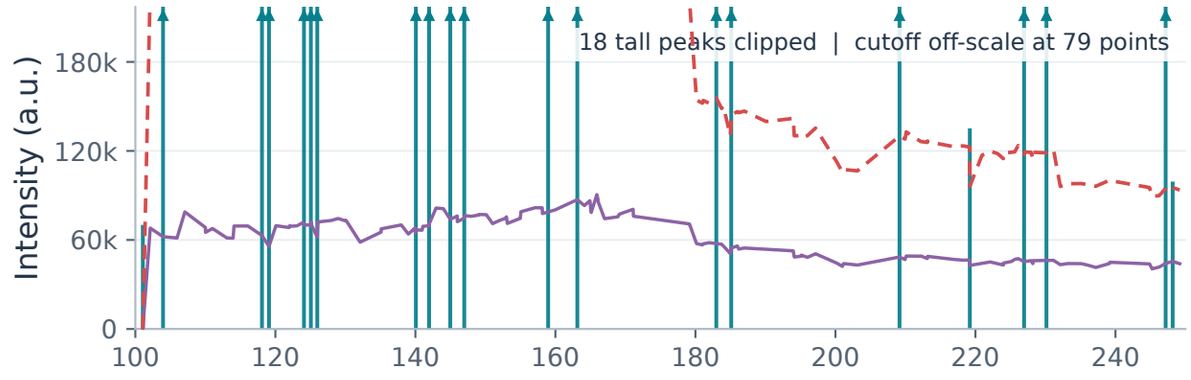
Background zoom

Median estimated background = 62.0k; 29% of axis height

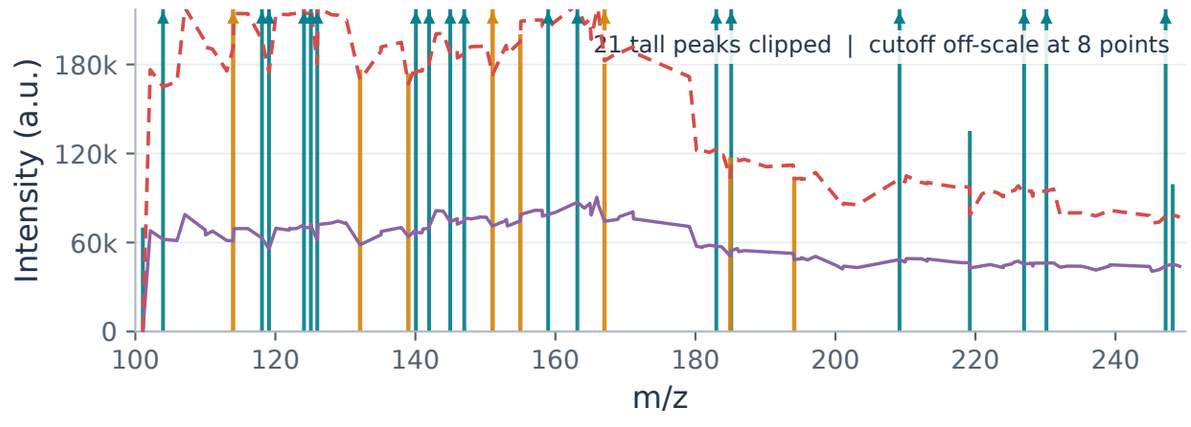
Before filtering · 148 centroids



Default 3σ · 21/148 kept (14.2%)



Less strict 2σ · 29/148 kept (+8)



Original centroids
Retained at 3σ
Additional at 2σ
Estimated background μ
Active cutoff μ + kσ

Same raw intensities in both columns; only the y-axis limits change. ▲ marks peaks extending above the axis.

Thresholds use the complete scan: 25 Da window, 2 clipping passes at 2σ, minimum 8 neighbors. Removed ≠ confirmed noise.